Mathematics 101

Integrals, Vectors, Sequences, and Trigonometry

Allowed aids: Calculator, writing materials, and formula sheet.

Task 1

Compute the integrals.

a) $\int_0^\pi x \cos x \, dx$

Write your solution in this box

b) $\int \frac{1}{x^2 + x} \, dx$

Write your solution in this box

Task 2

Determine t such that

$$\int_1^t \frac{1}{x+1} \, dx = \ln\left(\frac{5}{2}\right)$$

Task 3

The functions f and g are given by $f(x) = 4 - x^2$ and $g(x) = x + 2$. The graphs of f and g bound a region. What is the area of this region?

Task 4

The sum of the first ten terms of an arithmetic sequence is 90.

Determine the common difference d, given that $a_1 = 3$.

Task 5

The points $A(1, 2, 0)$, $B(4, 6, 2)$, and $C(8, 6, 4)$ are given.

a) Determine the area of $\triangle ABC$.

b) Find the distance from point C to the line through A and B.

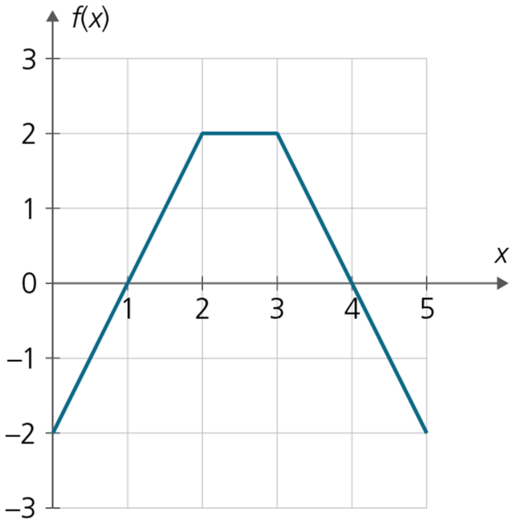
Task 6

Given the planes $\alpha : x + y + z = 6$ and $\beta : 2x - y + z = 3$.

Find a parametric equation for the line of intersection between the two planes.

Task 7

The figure below shows the graph of the function f.



The function g is given by $g(x) = \int_0^x f(t)\,dt$.

Determine $g(4)$ and $g'(4)$.

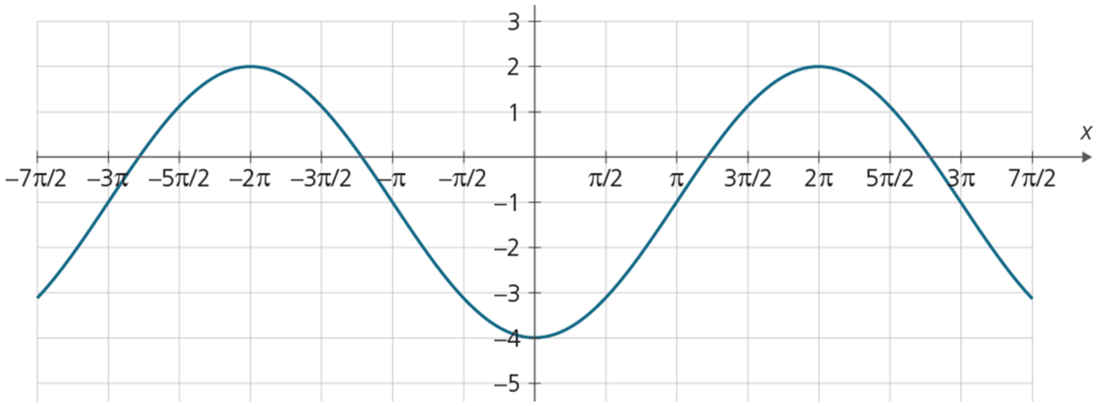
Task 8

An angle θ in the second quadrant is given by $\sin\theta = \frac{3}{5}$.

a) Determine $\cos\theta$ and $\tan\theta$.

b) Solve the equation $2\cos(2x) = 1$ for $x \in [0, 2\pi]$.

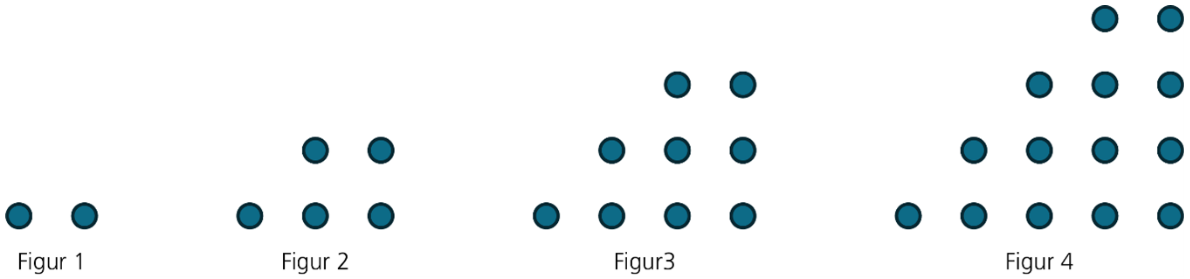
The figure below shows the graph of a function f.



c) Find an expression for f.

Task 9

Each figure below consists of dots arranged in a specific pattern.



Let F_n be the number of dots in figure n. The first four figure numbers are 2, 5, 9, and 14.

a) Determine F_6.

b) It can be shown that $F_n = F_{n-1} + n + 1$. Carry out an induction proof showing that $F_n = \dfrac{1}{2}n^2 + \dfrac{3}{2}n$ for all $n \geq 1$.

Task 10

Aisha is saving for a down payment to buy a home. On the 1st of each month, she transfers NOK 12,000 to a savings account with 0.3% monthly interest. The first transfer was on 1 January 2023.

a) Explain why the very first deposit has been in the account for 36 months by the end of December 2025.

b) How much does she have in the account by the end of December 2025?

c) How much would she need to deposit each month for the account balance to be half a million NOK by the end of December 2025?